DISCRETE MATHEMATICS AND ITS APPLICATIONS



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Course description

Chapter 1. The foundations: Logic and Proofs

Chapter 2. Basic Structures: Sets, Function, Sequences, and Sums

Chapter 3. The fundamentals: Algorithms and the Integers

Chapter 4. Induction and Recursion

Chapter (5+7). Counting & Advanced Counting Techniques

Chapter 9. Graphs

Chapter 10. Trees

REFERENCES

Student Textbook: Kenneth Rosen,
Discrete Mathematics and its applications. Ed.7 (or 8), Mc.Graw Hill, IE, 2007.

Evaluation Strategy

- Must attend more than 80% of contact hours
- Evaluating
  - 03 Progress tests: → 30%
  - 03 Assignments: → 30%
  - Final Exam (FE): → 40%
- Pass: Every on-going assessment component > 0
  and Final Exam score ≥ 4 (of 10)
  and Final Result ≥ 5
  Retake only the Final Exam when not passed

Chapter 1
The Foundations: Logic and Proofs

Contents

1.1-Propositional Logic – Logic mệnh đề
1.2-Propositonal Equivalences
1.3-Predicates and Quantifiers (vị từ và lượng từ)
1.4-Nested Quantifiers
1.5-Rules of Inference – Các quy tắc suy diễn

1.1- Propositional Logic

1.1.1- Definitions and Truth Table

1.1.2- Precedence of Logical Operators

1.1.1- Definitions and Truth Table

- **Proposition** is a declarative sentence that is either ***true*** or ***false*** but ***not both***.
- Proposition is a sentence that declares a ***fact***.
- **Examples 1**:
  * I am a girl
  * Ha Noi is not the capital of Vietnam
  * 1+5 < 4
  * What time is it?
  * X+Y=Z

OK

No OK

1.1.1- Definitions…

- **Truth table**
  - I am a girl

p
True/ T / 1
False / F / 0

1.1.1- Definitions…

- ***Negation*** of propositions p is the statement "It is not case that p".
- Notation: ¬p (or $\overline{p}$)

p	$\overline{p}$
1	0
0	1

1.1.1- Definitions…

EX: Find the negation of the proposition "Today is Friday."

1.1.1- Definitions…

EX: Find the negation of the proposition "Today is Friday."

Solution:
The negation is "It is not the case that today is Friday."

"Today is not Friday,"
or
"It is not Friday today."

1.1.1- Definitions…

- ***Conjunction*** of propositions p and q is the proposition "p and q" and denoted by p^q

p	q	p^q
0	0	0
0	1	0
1	0	0
1	1	1

1.1.1- Definitions…

Ex: Find the conjunction of the propositions p and q where p is the proposition "*Today is Friday*" and q is the proposition "*It is raining today*".

1.1.1- Definitions…

Ex: Find the conjunction of the propositions p and q where p is the proposition "*Today is Friday*" and q is the proposition "*It is raining today*".

Solution: The conjunction of these propositions, p^q , is the proposition ***"Today is Friday and it is raining today"***

This proposition is true on rainy Fridays and is false on any day that is not a Friday and on Fridays when it does not rain.

1.1.1- Definitions…

- Disjunction of propositions p and q is the proposition " p or q" and denoted by p ***v*** q

p	q	p?q
0	0	0
0	1	1
1	0	1
1	1	1

1.1.1- Definitions…

Ex: Find the disjunction of the propositions p and q where p is the proposition "Today is Friday" and q is the proposition "It is raining today"

1.1.1- Definitions…

Ex: Find the disjunction of the propositions p and q where p is the proposition "Today is Friday" and q is the proposition "It is raining today"

Solution: The disjunction of p and q , p v q, is the proposition ***"Today is Friday or it is raining today"***

This proposition is true on any day that is either a Friday or a rainy day (including rainy Fridays). It is only false on days that are not Fridays when it also does not rain.

1.1.1- Definitions…

- **Exclusive-or** (xor) of propositions p and q, denoted by $p \oplus q$

p	q	$p \oplus q$
0	0	0
0	1	1
1	0	1
1	1	0

1.1.1- Definitions…

- ***Implication***: $p \rightarrow q$ (p implies q)
- p: *hypothesis / antecedent / premise*
- q: *conclusion / consequence*
- *$p \rightarrow q$ can be expressed as:*
  - *q if p*
  - *If p, then q*
  - *p is sufficient condition for q*
  - *q is necessary condition for p*

p	q	p → q
0	0	1
0	1	1
1	0	0
1	1	1

"If 1 + 1 = 3, then dogs can fly"
→ TRUE
(p ? q)
p=0, q=0 ,
so (p?q) is true.

1.1.1- Definitions…

- **Biconditional statement** p ⍰ q is the proposition " p if and only if q"
- p → q (p ***only if*** q) and p⍰q (p ***if*** q)

p	q	p→q	q→p	(p→q) ^ (q→p)	p ↔ q
0	0	1	1	1	1
0	1	1	0	0	0
1	0	0	1	0	0
1	1	1	1	1	1

1.1.2- Precedence of Logical Operators

(1) Parentheses from inner to outer

(2) ¬

(3) ^

(4) ∨

(5) →

(6) ↔

Examples.

- ¬p ∧ q means (¬p) ∧ q, not ¬(p ∧ q)
- p ∧ q ∨ r means (p ∧ q) ∨ r, not p ∧ (q ∨ r)
- p ∨ q → r means (p ∨ q) → r, not p ∨ (q → r)

Exercises 8P15

8. Let p and q be the propositions

p : I bought a lottery ticket this week.

q : I won the million dollar jackpot.

Express each of these propositions as an English sentence.

a) $\neg p$ **b)** $p \vee q$ **c)** $p \rightarrow q$

d) $p \wedge q$ **e)** $p \leftrightarrow q$ **f)** $\neg p \rightarrow \neg q$

g) $\neg p \wedge \neg q$ **h)** $\neg p \vee (p \wedge q)$

Exercises 8P15

(a) I did not buy a lottery ticket this week.
(b) I bought a lottery ticket this week or I won the million dollar jackpot.
(c) If I bought a lottery ticket this week, then I won the million dollar jackpot.
(d) I bought a lottery ticket this week and I won the million dollar jackpot.
(e) I bought a lottery ticket this week if and only if I won the million dollar jackpot.
(f) If I did not buy a lottery ticket this week, then I did not win the million dollar jackpot.
(g) I did not buy a lottery ticket this week and I did not win the million dollar jackpot.
(h) I did not buy a lottery ticket this week or, I bought a lottery ticket this week and won the million dollar jackpot.

Exercises 13P15

Let p and q be the propositions

p : You drive over 65 miles per hour.
q : You get a speeding ticket.

Write these propositions using p and q and logical connectives (including negations).

a) You do not drive over 65 miles per hour.
b) You drive over 65 miles per hour, but you do not get a speeding ticket.
c) You will get a speeding ticket if you drive over 65 miles per hour.
d) If you do not drive over 65 miles per hour, then you will not get a speeding ticket.
e) Driving over 65 miles per hour is sufficient for getting a speeding ticket.
f) You get a speeding ticket, but you do not drive over 65 miles per hour.
g) Whenever you get a speeding ticket, you are driving over 65 miles per hour.

Exercises 13P15

(a) $\neg p$

(b) $p \wedge \neg q$

(c) $p \rightarrow q$

(d) $\neg p \rightarrow \neg q$

(e) $p \rightarrow q$

(f) $q \wedge \neg p$

(g) $q \rightarrow p$

Exercises

31. Construct a truth table for each of these compound propositions.

a) $p \wedge \neg p$
b) $p \vee \neg p$
c) $(p \vee \neg q) \rightarrow q$
d) $(p \vee q) \rightarrow (p \wedge q)$
e) $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p)$
f) $(p \rightarrow q) \rightarrow (q \rightarrow p)$

33. Construct a truth table for each of these compound propositions.

a) $(p \vee q) \rightarrow (p \oplus q)$
b) $(p \oplus q) \rightarrow (p \wedge q)$
c) $(p \vee q) \oplus (p \wedge q)$
d) $(p \leftrightarrow q) \oplus (\neg p \leftrightarrow q)$
e) $(p \leftrightarrow q) \oplus (\neg p \leftrightarrow \neg r)$
f) $(p \oplus q) \rightarrow (p \oplus \neg q)$

1.2- Propositional Equivalences

1.2.1- Tautology and Contradiction

1.2.2- Logical Equivalences

1.2.3. De Morga's Laws

1.2.1- Tautology and Contradiction

- *Tautology is a proposition that is **always true***
- *Contradiction is a proposition that is **always false***
- When p $\leftrightarrow$ q is tautology, we say "p and q are called ***logically equivalence***". Notation: p $\equiv$ q

Example 3 p.23

- *Show that* $p \leftrightarrow q$ *and* $\neg p \vee q$ *are logically equivalent.*

TABLE 4 Truth Tables for $\neg p \vee q$ and $p \rightarrow q$.

p	q	$\neg p$	$\neg p \vee q$	$p \rightarrow q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

1.2.2- Logical Equivalences

Equivalence	Name
$p \wedge T \equiv p$ $\quad p \vee F \equiv p$	Identity laws- Luật đồng nhất
$p \vee T \equiv T$ $\quad p \wedge F \equiv F$	Domination Laws – Luật chi phối
$p \vee p \equiv p$ $\quad p \wedge p \equiv p$	Idempotent Laws – Luật bất biến
$\neg(\neg p) \equiv p$	Double Negation Laws – Luật đảo kép
$p \vee q \equiv q \vee p$ $\quad p \wedge q \equiv q \wedge p$	Commutative Laws – Luật giao hoán
$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative Laws – Luật kết hợp
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive Laws – Luật phân phối
$\neg (p \wedge q) \equiv \neg p \vee \neg q$ $\quad \neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan Laws
$p \vee (p \wedge q) \equiv p$ $\quad p \wedge (p \vee q) \equiv p$	Absorption Laws – Luật hấp thụ
$p \vee \neg p \equiv T$ $\quad p \wedge \neg p \equiv F$	Negation Laws - Luật nghịch đảo

1.2.2- Logical Equivalences

Equivalences	Equivalences
$p \rightarrow q \equiv \neg p \vee q$	$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$
$p \rightarrow q \equiv \neg q \rightarrow \neg p$	$p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$
$p \vee q \equiv \neg p \rightarrow q$	$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$
$p \wedge q \equiv \neg (p \rightarrow \neg q)$	$\neg (p \leftrightarrow q) \equiv p \leftrightarrow \neg q$
$\neg(p \rightarrow q) \equiv p \wedge \neg q$	
$(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$	
$(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$	
$(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$	
$(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$	

1.2.2- Logical Equivalences

Example: Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent

Solution:

$$\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg(\neg p \wedge q)$$
$$\equiv \neg p \wedge [\neg(\neg p) \vee \neg q]$$
$$\equiv \neg p \wedge (p \vee \neg q)$$
$$\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q)$$
$$\equiv \mathbf{F} \vee (\neg p \wedge \neg q)$$
$$\equiv (\neg p \wedge \neg q) \vee \mathbf{F}$$
$$\equiv \neg p \wedge \neg q$$

Consequently $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent.

1.2.2- Logical Equivalences

Example: Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology

Solution:

$$(p \wedge q) \rightarrow (p \vee q) \equiv \neg(p \wedge q) \vee (p \vee q)$$
$$\equiv (\neg p \vee \neg q) \vee (p \vee q)$$
$$\equiv (\neg p \vee p) \vee (\neg q \vee q)$$
$$\equiv \mathbf{T} \vee \mathbf{T}$$
$$\equiv \mathbf{T}$$

1.2.2- Logical Equivalences

Equivalence	Name
$p \wedge T \equiv p$ $\quad p \vee F \equiv p$	Identity laws- Luật đồng nhất
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$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative Laws – Luật kết hợp
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive Laws – Luật phân phối
$\neg (p \wedge q) \equiv \neg p \vee \neg q$ $\quad \neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan Laws
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$p \vee \neg p \equiv T$ $\quad p \wedge \neg p \equiv F$	Negation Laws - Luật nghịch đảo

1.2.2- Logical Equivalences

Equivalences	Equivalences
$p \rightarrow q \equiv \neg p \vee q$	$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$
$p \rightarrow q \equiv \neg q \rightarrow \neg p$	$p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$
$p \vee q \equiv \neg p \rightarrow q$	$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$
$p \wedge q \equiv \neg (p \rightarrow \neg q)$	$\neg (p \leftrightarrow q) \equiv p \leftrightarrow \neg q$
$\neg(p \rightarrow q) \equiv p \wedge \neg q$	
$(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$	
$(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$	
$(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$	
$(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$	

Exercises

16. Determine whether these biconditionals are true or false.

a) 2 + 2 = 4 if and only if 1 + 1 = 2.

b) 1 + 1 = 2 if and only if 2 + 3 = 4.

c) 1 + 1 = 3 if and only if monkeys can fly.

d) 0 > 1 if and only if 2 > 1.

1.3- Predicates and Quantifiers

- Introduction
- Predicates
- Quantifiers

1.3.1- Introduction

A type of logic used to express the meaning of a wide range of statements in mathematics and computer science in ways that permit us to reason and explore relationships between objects.

1.3.2- Predicates – vị từ

- X > 0
- P(X)="X is a prime number" (số nguyên tố), called propositional function at X.
- P(2)="2 is a prime number" ≡True
- P(4)="4 is a prime number" ≡False

1.3.2- Predicates – vị từ

- $Q(X_1,X_2,\ldots,X_n)$, n-place/ n-ary predicate
- Example: "x=y+3" ➔ Q(x,y)

  Q(1,2) ≡ "1=2+3" ≡ false

  Q(5,2) ≡ "5=2+3" ≡ true

1.3.2- Predicates

- Predicates are pre-conditions and post-conditions of a program.
- If x>0 then x:=x+1
  - Predicate: "x>0" ➔ P(x)
  - Pre-condition: P(x)
  - Post-condition: P(x)
- T:=X;
  X:=Y;
  Y:=T;
    - Pre-condition: "x=a and y=b" ➔ P(x, y)
    - Post-condition: "x=b and y=a" ➔ Q(x, y)

Pre-condition (P(...)) : condition describes valid input.
Post-condition (Q(...)) : condition describe valid output of the codes.
Show the verification that a program always produces the desired output:
P(...) is true
Executing Step 1.
Executing Step 2.
.....
Q(...) is true

1.3.3- Quantifiers – Lượng từ

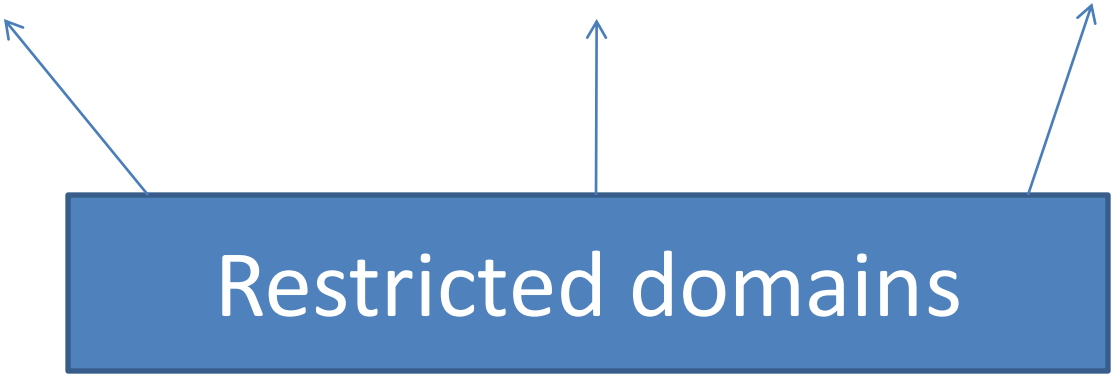
- The words in natural language: all, some, many, none, few, ….are used in quantifications.
- Predicate Calculus: area of logic that deals with predicates and quantifiers.
- The ***universal quantification*** (lượng từ phổ dụng) *of P(x) is the statement "P(x) for all values of x in the domain".* Notation : $\forall xP(x)$
- The ***existential quantification*** (lượng từ tồn tại) *of P(x) is the statement "There exists an element x in the domain such that P(x)".* Notation : $\exists xP(x)$
- **Uniqueness quantifier**: $\exists !x\ P(x)$ or $\exists_1 xP(x)$
- $\forall xP(x) \vee Q(y)$:
  - x is a bound variable
  - y is a free variable

1.3.4- Quantifiers with Restricted Domains

$\forall x<0(x^2 > 0), \forall y \neq 0(y^3 \neq 0), \exists z>0(z^2 =2)$

$\leftrightarrow$

$\forall x(X<0 \wedge x^2 > 0), \forall y(y \neq 0 \wedge y^3 \neq 0), \exists z(z>0 \wedge z^2 =2)$



1.3.5- Precedence of Quantifiers

- Quantifier have higher precedence than all logical operators from propositional calculus.
- $\forall xP(x) \vee Q(x) \rightarrow (\forall xP(x)) \vee Q(x)$
- $\forall$ has higher precedence. So, $\forall$ affects on P(x) only.

1.3.6- Logical Equivalences Involving Quantifiers

Statements involving predicates and quantifiers are ***logically equivalent if and only if they have the same truth value*** no matter which predicates are substituted into the statements and which domain of discourse is used for the variables in these propositional functions.

- $\forall x\,(P(x) \wedge Q(x)) \equiv \forall x P(x) \wedge \forall x Q(x)$
  - Proof: page 43

Expression	Equivalence	Expression	Negation
$\neg\exists x P(x)$	$\forall x \neg P(x)$	$\exists x P(x)$	$\forall x \neg P(x)$
$\neg \forall x P(x)$	$\exists x \neg P(x)$	$\forall x P(x)$	$\exists x \neg P(x)$

1.3.7- Translating

- For every student in the class has studied calculus
- For every student in the class, that student has studied calculus
- For every student x in the class, x has studied calculus
- $\forall x\,(S(x) \rightarrow C(x))$

Exercises

1. Let $P(x)$ be the statement "$x = x^2$." If the domain consists of the integers, what are these truth values?

a) $P(0)$ **b)** $P(1)$ **c)** $P(2)$

d) $P(-1)$ **e)** $\exists x P(x)$ **f)** $\forall x P(x)$

2. Suppose that the domain of $Q(x, y, z)$ consists of triples x, y, z, where $x = 0, 1,$ or 2, $y = 0$ or 1, and $z = 0$ or 1. Write out these propositions using disjunctions and conjunctions.

a) $\forall y Q(0, y, 0)$ **b)** $\exists x Q(x, 1, 1)$

c) $\exists z \neg Q(0, 0, z)$ **d)** $\exists x \neg Q(x, 0, 1)$

a) $Q(0,0,0) \wedge Q(0,1,0)$
b) $Q(0,1,1) \vee Q(1,1,1) \vee Q(2,1,1)$
c) $\neg Q(0,0,0) \vee \neg Q(0,0,1)$
d) $\neg Q(0,0,1) \vee \neg Q(1,0,1) \vee \neg Q(2,0,1)$

Exercises 8, P53

Translate these statements into English, where $R(x)$ is "x is a rabbit" and $H(x)$ is "x hops" and the domain consists of all animals.

a) $\forall x(R(x) \rightarrow H(x))$

b) $\forall x(R(x) \wedge H(x))$

c) $\exists x(R(x) \rightarrow H(x))$

d) $\exists x(R(x) \wedge H(x))$

a) All rabbits hop.

b) All animals are hopping rabbits.

c) There exists an animal that, if it is a rabbit, then it hops.

d) Some rabbits hop.

Example

Translate these statements into English, where $C(x)$ is "x is a comedian" and $F(x)$ is "x is funny" and the domain consists of all people.

a) $\forall x(C(x) \rightarrow F(x))$

b) $\forall x(C(x) \wedge F(x))$

c) $\exists x(C(x) \rightarrow F(x))$

d) $\exists x(C(x) \wedge F(x))$

A) Every comedian is funny
B) Every people are funny comedian
C) There exist, people, that, if it is a comedian, then they funny
D) Some comedians are funny

Example

Translate these statements into English, where $C(x)$ is "x is a comedian" and $F(x)$ is "x is funny" and the domain consists of all people.

a) $\forall x(C(x) \rightarrow F(x))$

b) $\forall x(C(x) \wedge F(x))$

c) $\exists x(C(x) \rightarrow F(x))$

d) $\exists x(C(x) \wedge F(x))$

a) Every comedian is funny.

b) Every person is a funny comedian.

c) There exists a person such that if she or he is a comedian, then she or he is funny.

d) Some comedians are funny.

Example

Let $P(x)$ be the statement "x can speak Russian" and let $Q(x)$ be the statement "x knows the computer language C++." Express each of these sentences in terms of $P(x)$, $Q(x)$, quantifiers, and logical connectives. The domain for quantifiers consists of all students at your school.

a) There is a student at your school who can speak Russian and who knows C++.

b) There is a student at your school who can speak Russian but who doesn't know C++.

c) Every student at your school either can speak Russian or knows C++.

d) No student at your school can speak Russian or knows C++.

a) $\exists x(P(x) \wedge Q(x))$

b) $\exists x(P(x) \wedge \neg Q(x))$

c) $\forall x(P(x) \vee Q(x))$

d) $\forall x \neg(P(x) \vee Q(x))$

Example

Let *C(x)* be the statement "*x* has a cat," let *D(x)* be the statement "*x* has a dog," and let *F(x)* be the statement "*x* has a ferret." Express each of these statements in terms of *C(x)*, *D(x)*, *F(x)*, quantifiers, and logical connectives. Let the domain consist of all students in your class.

a) A student in your class has a cat, a dog, and a ferret.

b) All students in your class have a cat, a dog, or a ferret.

c) Some student in your class has a cat and a ferret, but not a dog.

d) No student in your class has a cat, a dog, and a ferret.

e) For each of the three animals, cats, dogs, and ferrets, there is a student in your class who has this animal as a pet

a) $\exists x(C(x) \wedge D(x) \wedge F(x))$ **b)** $\forall x(C(x) \vee D(x) \vee F(x))$

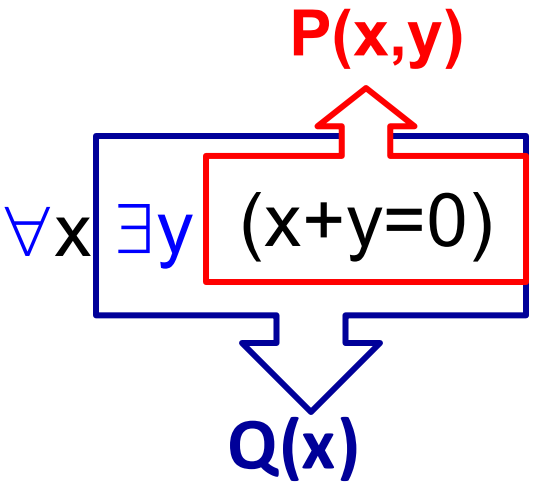
c) $\exists x(C(x) \wedge F(x) \wedge \neg D(x))$ **d)** $\neg\exists x(C(x) \wedge Q(x) \wedge F(x))$

e) $(\exists x(C(x)) \wedge (\exists x D(x)) \wedge (\exists x F(x))$

1.4. Nested Quantifiers

Two quantifiers are nested if one is within the scope of the other

Example:



$\forall x\, Q(x)$

$Q(x)$ is $\exists y\, P(x,y)$

$P(x,y)$ is $(x+y=0)$

Nested Quantifiers

$\forall x \exists y\ Q(x, y)$ vs $\exists y \forall x\ Q(x, y)$

THE SAME???

NOT OK

How to understand **nested quantifiers**?

- Example 1.

x, y in domain is {1, 2, 3}

Q(x, y) = “x + y = 4”

➔ $\forall x \exists y Q(x, y)$

How to understand **nested quantifiers**?

- Example 1.

x, y in domain is {1, 2, 3}

Q(x, y) = "x + y = 4"

➔ $\forall x \exists y Q(x, y)$

$\equiv \exists y Q(1,y) \wedge \exists y Q(2,y) \wedge \exists y Q(3,y)$

$\equiv \exists y$ "1 + y = 4" $\wedge \exists y$ "2 + y = 4" $\wedge \exists y$ "3 + y = 4"

➔ $T \wedge T \wedge T = T$

➔ TRUE

How to understand **nested quantifiers**?

- Example 2.

x, y in domain is {1, 2, 3}

Q(x, y) = “x + y = 4”

➔ $\exists x \forall y Q(x, y)$

How to understand **nested quantifiers**?

- Example 2.

x, y in domain is {1, 2, 3}

Q(x, y) = "x + y = 4"

➔ $\exists x \forall y Q(x, y)$

$\equiv \forall y Q(1,y) \vee \forall y Q(2,y) \vee \forall y Q(3,y)$

➔ $\forall y(1+y=4) \vee \forall y(2+y=4) \vee \forall y(3+y=4)$

➔ $F \vee F \vee F$

➔ FALSE

How to negate a statement?

- Use De Morgan's laws
- **Example 1:** For x in domain {1, 2, 3}

$\neg \forall x P(x)$

$\equiv \neg(\forall x P(x))$

$\equiv \neg((P(1) \wedge P(2) \wedge P(3))$

$\equiv \neg P(1) \vee \neg P(2) \vee \neg P(3)$

$\equiv \exists x \neg P(x)$

How to negate a statement? **Example 2**

x: students in your class

P(x) = "x has taken a course in calculus"

"Every student in your class has taken a course in calculus"

➔ $\forall x\, P(x)$

Find the negation:

$\neg \forall x\, P(x) \equiv \exists x \neg P(x)$

➔ "There is a student in your class who has not taken a course in calculus."

How to negate a statement?

- Use De Morgan's laws
- **Example 3:** For x in domain {1, 2, 3}

$\neg\exists x P(x)$

$\equiv \neg(\exists x P(x))$

$\equiv \neg((P(1) \vee P(2) \vee P(3))$

$\equiv \neg P(1) \wedge \neg P(2) \wedge \neg P(3)$

$\equiv \forall x \neg P(x)$

How to negate a statement?

Example 4

- Find the negation of the statement

$\exists x(x^2 = 2)$.

Solution.

$\neg\exists x(x^2 = 2) \equiv \forall x\neg(x^2 = 2) \equiv \forall x(x^2 \neq 2)$

➔The truth value of this statement depends on the domain.

-If x is an integer: $\exists x(x^2 = 2)$is FALSE and $\forall x(x^2 \neq 2)$ is TRUE

-If x is a real number: $\exists x(x^2 = 2)$is TRUE and $\forall x(x^2 \neq 2)$ is FALSE

How to negate a statement?

- Use De Morgan's laws
- **Example 5:** For x, y in domain {1, 2, 3}

$\neg\forall x\exists y Q(x,y)$

$\equiv \neg(\exists y Q(1,y) \wedge \exists y Q(2,y) \wedge \exists y Q(3,y))$

$\equiv \neg\exists y Q(1,y) \vee \neg\exists y Q(2,y) \vee \neg\exists y Q(3,y)$

$\equiv \forall y\neg Q(1,y) \vee \forall y\neg Q(2,y) \vee \forall y\neg Q(3,y)$

$\equiv \exists x\forall y\neg Q(x,y)$

How to negate a statement? Example 6

- Find the negation of the statement

"Every student in this class plays some sport."

Solution.

x: student in this class

y: sport

P(x, y) = "student x ***plays*** sport y"

"Every student in this class **plays** some sport."

➔ $\forall x \exists y P(x,y)$

➔ Negation: $\neg \forall x \exists y P(x,y) \equiv \exists x \forall y \neg P(x,y)$

➔ "There are some students in this class who don't play any sport.

Find the negations

- $\forall x\,(P(x) \rightarrow Q(x))$

Negation:

$\neg\forall x\,(P(x) \rightarrow Q(x))$

$\equiv \exists x \neg\,(P(x) \rightarrow Q(x))$

$\equiv \exists x\,(P(x) \wedge \neg Q(x))$

- $\forall x\,(P(x) \wedge Q(x))$

Negation:

$\neg\forall x\,(P(x) \wedge Q(x))$

$\equiv \exists x \neg\,(P(x) \wedge Q(x))$

$\equiv \exists x\,(\neg P(x) \vee \neg Q(x))$

How to **translate** a statement into logical expression?

- **Example 1.** Suppose the variables ***x*** represents **real numbers**, and

$E(x)$: x is even $I(x)$: x is an integer.

Write the statement "Every integer is even" using these predicates and any needed quantifiers.

Ans. "Every integer is even " means

"For every number, if it is an integer, then it is even"

➔"For every **x**, **if** **x** is an integer, **then** **x** is even"

$\forall x \quad (\quad I(x) \quad \rightarrow \quad E(x) \quad)$

$$\forall x \, (I(x) \rightarrow E(x))$$

How to **translate** a statement into logical expression?

Example 2. x : represents people

$F(x)$: x is friendly; $T(x)$: x is tall; $A(x)$: x is angry.

- Bill is angry.

$$A(Bill)$$

- Some people are not angry.

$$\exists x \neg A(x)$$

- All tall people are friendly.

➔ For every person x, if x is tall, then x is friendly. $\forall x\ (T(x) \rightarrow F(x))$

How to **translate** a statement into logical expression?

Example 3. *x:* represents students; *y* represents courses, and

$T(x, y)$: x is taking y $P(x, y)$: x passed y.

Write the statement in good English. Do not use variables in your answers.

- $\forall x \exists y T(x, y)$

Every student is taking at least one course.

- $\exists y \forall x T(x, y)$

There is a course that all students are taking.

- $\forall x \exists y \neg P(x, y)$

Every student didn't pass at least one course.

Do yourself. Translating

Example 4. Assume that the universe for x is all people and the universe for y is the set of all movies.

$S(x, y)$: x saw y $L(x, y)$: x liked y $A(y)$: y won an award $C(y)$: y is a comedy.

- No comedy won an award.
- $S(\text{Lois, Fast 8}) \wedge \neg L(\text{Lois, Fast 8})$
- Some people have seen every comedy.
- $\neg\exists x \forall y(S(x, y) \rightarrow L(x, y))$
- Ben has never seen a movie that won an award.

Do yourself. Translating

Example 4. Assume that the universe for x is all people and the universe for y is the set of all movies.

$S(x, y)$: x saw y $L(x, y)$: x liked y $A(y)$: y won an award $C(y)$: y is a comedy.

1. No comedy won an award.
2. $S(\text{Lois, Fast 8}) \wedge \neg L(\text{Lois, Fast 8})$
3. Some people have seen every comedy.
4. $\neg\exists x\forall y(S(x, y) \rightarrow L(x, y))$
5. Ben has never seen a movie that won an award.

Answers:

1. $\forall y(C(y) \rightarrow \neg A(y))$
2. Lois saw *Fast 8*, but didn't like it.
3. $\exists x\forall y(C(y) \rightarrow S(x, y))$
4. No one liked every movie he has seen.
5. $\neg\exists y(S(\text{Ben}, y)\wedge A(y))$

The order of quantifiers

TABLE 1 Quantifications of Two Variables.

Statement	*When True?*	*When False?*
$\forall x \forall y P(x, y)$ $\forall y \forall x P(x, y)$	$P(x, y)$ is true for every pair x, y.	There is a pair x, y for which $P(x, y)$ is false.
$\forall x \exists y P(x, y)$	For every x there is a y for which $P(x, y)$ is true.	There is an x such that $P(x, y)$ is false for every y.
$\exists x \forall y P(x, y)$	There is an x for which $P(x, y)$ is true for every y.	For every x there is a y for which $P(x, y)$ is false.
$\exists x \exists y P(x, y)$ $\exists y \exists x P(x, y)$	There is a pair x, y for which $P(x, y)$ is true.	$P(x, y)$ is false for every pair x, y.

Exercises

Let $Q(x, y)$ be the statement "$x + y = x - y$." If the domain for both variables consists of all integers, what are the truth values?

a) $Q(1, 1)$
b) $Q(2, 0)$
c) $\forall y Q(1, y)$
d) $\exists x Q(x, 2)$
e) $\exists x \exists y Q(x, y)$
f) $\forall x \exists y Q(x, y)$
g) $\exists y \forall x Q(x, y)$
h) $\forall y \exists x Q(x, y)$
i) $\forall x \forall y Q(x, y)$

Exercises

Let *L(x, y)* be the statement "*x* loves *y*," where the domain for both *x* and *y* consists of all people in the world.

Use quantifiers to express each of these statements.

a) Everybody loves Jerry.

b) Everybody loves somebody.

c) There is somebody whom everybody loves.

d) Nobody loves everybody.

e) There is somebody whom Lydia does not love.

f) There is somebody whom no one loves.

g) There is exactly one person whom everybody loves.

h) There are exactly two people whom Lynn loves.

i) Everyone loves himself or herself.

j) There is someone who loves no one besides himself or herself.

Exercises

a) $\forall x L(x, \text{Jerry})$

b) $\forall x \exists y L(x, y)$

c) $\exists y \forall x L(x, y)$

d) $\forall x \exists y \neg L(x, y)$

e) $\exists x \neg L(\text{Lydia}, x)$

f) $\exists x \forall y \neg L(y, x)$

g) $\exists x(\forall y L(y, x) \wedge \forall z((\forall w L(w, z)) \rightarrow z = x))$

h) $\exists x \exists y(x = y \wedge L(\text{Lynn}, x) \wedge L(\text{Lynn}, y) \wedge \forall z(L(\text{Lynn}, z) \rightarrow (z = x \vee z = y)))$

i) $\forall x L(x, x)$

j) $\exists x \forall y (L(x,y) \leftrightarrow x = y)$

Exercises

Let *S(x)* be the predicate "*x* is a student," *F (x)* the predicate "*x* is a faculty member," and *A(x, y)* the predicate "*x* has asked *y* a question," where the domain consists of
all people associated with your school. Use quantifiers to express each of these statements.

a) Lois has asked Professor Michaels a question.

b) Every student has asked Professor Gross a question.

c) Every faculty member has either asked Professor Miller a question or been asked a question by Professor Miller.

d) Some student has not asked any faculty member a question.

e) There is a faculty member who has never been asked a question by a student.

f) Some student has asked every faculty member a question.

g) There is a faculty member who has asked every other faculty member a question.

h) Some student has never been asked a question by a faculty member.

Exercises

a) A(Lois, Professor Michaels)

b) $\forall x(S(x) \rightarrow A(x,$ Professor Gross$))$

c) $\forall x(F(x) \rightarrow (A(x,$ Professor Miller$) \vee A($Professor Miller, $x)))$

d) $\exists x(S(x) \wedge \forall y(F(y) \rightarrow \neg A(x, y)))$

e) $\exists x(F(x) \wedge \forall y(S(y) \rightarrow \neg A(y,x)))$

f) $\forall y(F(y) \rightarrow \exists x(S(x) \vee A(x,y)))$

g) $\exists x(F(x) \wedge \forall y((F(y) \wedge (y = x)) \rightarrow A(x,y)))$

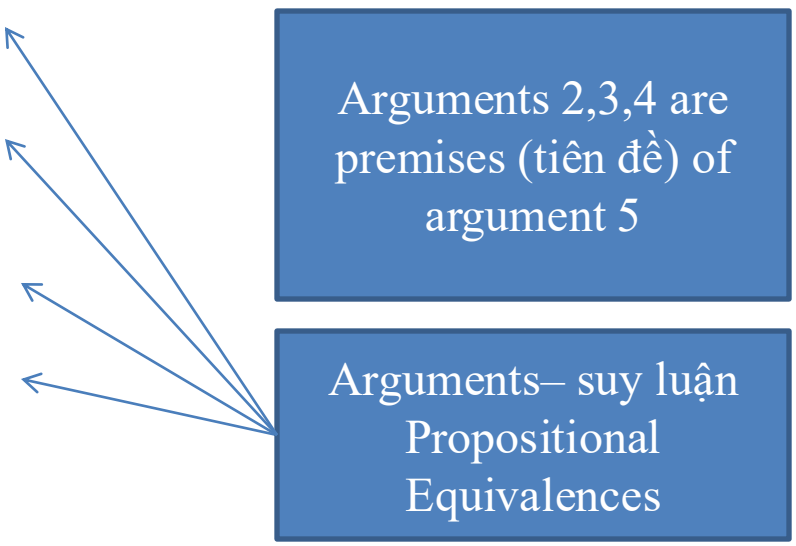
h) $\exists x(S(x) \wedge \forall y(F(y) \rightarrow \neg A(y, x)))$

1.5- Rules of Inference – Quy tắc diễn dịch

- Definitions
- Rules of Inferences

1.5.1- Definitions

- Proposition 1 // Hypothesis – giả thiết
- Proposition 2
- Proposition 3
- Proposition 4
- Proposition 5
- .....
- Conclusion



1.5.2- Rules Inferences

Rule	Tautology	Name
p p →q ∴ q	[p^ (p→q)] → q **You work hard** If **you work hard** then **you will pass the examination** **∴ you will pass the examination**	Modus ponen
¬q p → q ∴ ¬p	[¬q ^(p → q)] → ¬p **She did not get a prize** If **she is good at learning** **she will get a prize** **∴ She is not good at learning**	Modus tollen
p →q q →r ∴ p →r	[(p →q) ^(q →r)] →(p→r) If **it rains today**, then **we will not have a barbecue today**. If **we do not have a barbecue today**, then we **will have a barbecue tomorrow.** Therefore, if **it rains today**, then **we will have a barbecue tomorrow.**	Hypothetical syllolism – Tam đoạn luận giả thiết, Quy tắc bắc cầu

If Socrates is human, then Socrates is mortal.
Socrates is human.
∴ Socrates is mortal.

Một ngôi nhà rẻ thì hiếm
Cái gì hiếm thì đắt
∴ Một ngôi nhà rẻ thì đắt.

Rules Inferences…

Rule	Tautology	Name
$p \vee q$ $\neg p$ $\therefore q$	$[(p \vee q) \wedge \neg p] \rightarrow q$ **Power puts off** or **the lamp is malfunctional** **Power doesn't put off** **the lamp is malfunctional**	Disjunctive syllogism
p $\therefore p \vee q$	$p \rightarrow (p \vee q)$ **It is below freezing now** **It is below freezing now** or **raining now**	Addition

Rules Inferences…

Rule	Tautology	Name
$\frac{p \wedge q}{\therefore p}$	$(p \wedge q) \rightarrow p$ It is below freezing now and raining now It is below freezing now	Simplication
$\frac{p \quad q}{\therefore p \wedge q}$	$[(p) \wedge (q)) \rightarrow (p \wedge q)$	Conjunction
$\frac{p \vee q \quad \neg p \vee r}{\therefore q \vee r}$	$[(p \vee q) \wedge (\neg p \vee r)] \rightarrow (q \vee r)$ Jasmin is skiing OR it is not snowing It is snowing OR Bart is playing hockey Jasmin is skiing OR Bart is playing hockey	Resolution

1.5.3- Fallacies – ngụy biện – sai logic

- If **you do every problem in this book** then you will learn discrete mathematic

  You learned mathematic

  $(p \rightarrow q) \wedge q$

  $=(\neg p \vee q) \wedge q$

  (absorption law)

  $= q$

  ? No information for p

p	q	p->q	(p->q)^q
1	1	1	1
1	0	0	0
0	1	1	1
0	0	1	0

  p can be true or false ➔ You may learn discrete mathematic but you might do some problems only.

Fallacies…

- (p → q)^q □ p is not a tautology
  (it is false when p = 0, q = 1)
- (p □ q)^¬p □ ¬q is not a tautology
  (it is false when p = 0, q = 1)

Fallacies…

Hắn chửi như những người say rượu hát. Giá **hắn biết hát** thì hắn có lẽ **hắn không cần chửi**. Khổ cho hắn và khổ cho người, **hắn lại không biết hát**. Thì hắn chửi, cũng như chiều nay hắn chửi….. (Nam Cao, Chí Phèo, trang 78)

p → ¬ **q**

¬ **p**

➔ ¬(¬**q**) = **q** là không hợp logic

$$\frac{p \qquad p \rightarrow q}{\therefore q}$$

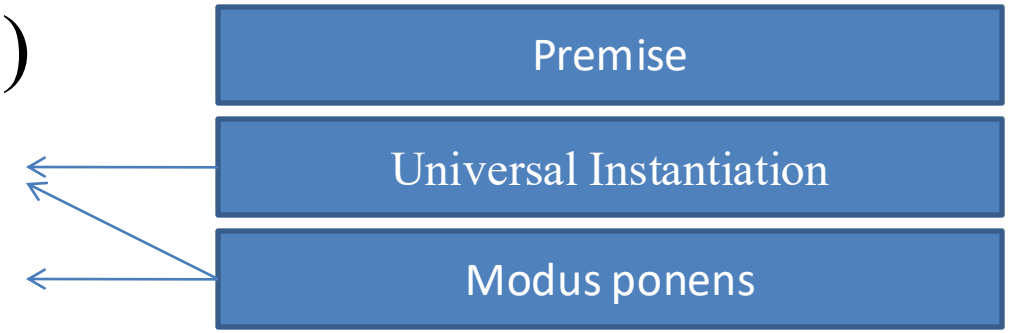
$$\frac{\neg q \qquad p \rightarrow q}{\therefore \neg p}$$

1.5.4- Rules of Inference for Quantified Statements

Rule	Name
$\forall x P(x)$ $\therefore P(c)$	Universal Instantiation Cụ thể hóa lượng từ phổ dụng
$P(c)$ for arbitrary c $\therefore \forall x P(x)$	Universal generalization Tổng quát hóa bằng lượng từ phổ dụng
$\exists x P(x)$ $\therefore P(c)$ for some element c	Existential instantiation Chuyên biệt hóa
$P(c)$ for some element c $\therefore \exists x P(x)$	Existential generalization Khái quát hóa bằng lượng từ tồn tại

Rules of Inference for Quantified Statements…

- "All student are in this class had taken the course PFC"
- "HB is in this class"
- "Had HB taken PFC?"
- $\forall x(P(x) \rightarrow Q(x))$
- $P(HB) \rightarrow Q(HB)$
- $P(HB)$
- $Q(HB)$ // conclusion



Summary

- Propositional Logic – Luận lý mệnh đề
- Propositional Equivalences
- Predicates and Quantifiers
- Nested Quantifiers
- Rules and Inference – Quy tắc và diễn dịch